## Theory Assignment

## Report: Finding the Largest Frame Size for Cyclic Schedulers

**Introduction & Standard Requirements** In a cyclic structured scheduler, the frame size (f) must satisfy three fundamental requirements to ensure a feasible schedule without preemptions within a frame:

1. **Requirement 1 (Execution Time):** The frame size must be large enough to execute the longest task without preemption. $f >= max(e_i)$
2. **Requirement 2 (Hyperperiod Divisibility):** The frame size must evenly divide the hyperperiod (H) to ensure the schedule repeats perfectly. $H \bmod f = 0$
3. **Requirement 3 (Deadline Satisfaction):** There must be at least one full frame between a task's release time and its deadline to guarantee completion. $2f - gcd(p_i, f) <= d_i$ (for all i)

Below is the theoretical analysis for each provided task set based on these requirements.

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### 1. Task Set 1 Analysis

**Tasks:** T1(15, 1, 14), T2(20, 2, 26), T3(22, 3) *Note: For T3, the implicit deadline is equal to its period, so d3 = 22.*

- **Hyperperiod (H):** LCM(15, 20, 22) = 660
- **Requirement 1:** f >= max(1, 2, 3) -> f >= 3
- **Requirement 2:** Candidates for f >= 3 that divide 660 are: 3, 4, 5, 6, 10, 11, 12...
- **Requirement 3 Check:** We test the largest candidates to find the maximum possible f.
  - **Testing f = 10:** For T1: 2(10) - gcd(15, 10) = 20 - 5 = 15. Since 15 is not <= 14, f = 10 is **invalid**.
  - **Testing f = 6:** For T1: 2(6) - gcd(15, 6) = 12 - 3 = 9 <= 14 (Valid) For T2: 2(6) - gcd(20, 6) = 12 - 2 = 10 <= 26 (Valid) For T3: 2(6) - gcd(22, 6) = 12 - 2 = 10 <= 22 (Valid)

**Conclusion for Set 1:** The largest possible frame size is **f = 6**.

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### 2. Task Set 2 Analysis

**Tasks:** T1(4, 1), T2(5, 2, 7), T3(20, 5) *Note: For T1 and T3, implicit deadlines equal their periods (d1 = 4, d3 = 20).*

- **Hyperperiod (H):** LCM(4, 5, 20) = 20
- **Requirement 1:** f >= max(1, 2, 5) -> f >= 5

- **Requirement 3 Check:** Let's apply Req 3 for T1 using the minimum possible frame size from Req 1 (f >= 5): Equation: 2f - gcd(4, f) <= 4 Since f >= 5, 2f will be at least 10. The maximum possible value for gcd(4, f) is 4. Therefore, the left side of the equation will always be >= 6 (which is 10 - 4). Since 6 is not <= 4, this requirement strictly fails for T1.

**Conclusion for Set 2: No valid frame size exists.** A frame must be at least 5 units long to accommodate T3, but T1 has a strict deadline of 4. In a pure cyclic scheduler (without applying "task slicing" to T3), this task set is unschedulable.

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**3. Task Set 3 Analysis**

**Tasks:** T1(5, 0.1), T2(7, 1), T3(12, 6), T4(45, 9) *Note: All deadlines are implicit (d1 = 5, d2 = 7, d3 = 12, d4 = 45).*

- **Hyperperiod (H):** LCM(5, 7, 12, 45) = 1260
- **Requirement 1:** f >= max(0.1, 1, 6, 9) -> f >= 9
- **Requirement 3 Check:** Let's apply Req 3 for T1 (p1 = 5, d1 = 5) using the constraint f >= 9: Equation: 2f - gcd(5, f) <= 5 Since f >= 9, 2f >= 18. The maximum possible value for gcd(5, f) is 5. Therefore, the left side is at least 13 (which is 18 - 5). Since 13 is not <= 5, Requirement 3 fails completely.

**Conclusion for Set 3: No valid frame size exists.** The execution time of T4 forces the frame size to be at least 9, which makes it mathematically impossible to guarantee T1's deadline of 5 without task slicing.

## SimSo Simulation Report: Rate Monotonic (RM) Scheduling

**1. What is the utilization factor of the system and what is the value for Urm(3)?**

- **System Utilization (U):** The utilization factor is calculated by summing the ratio of execution time to period for each task (U = e1/p1 + e2/p2 + e3/p3). U = (0.5 / 2) + (1.2 / 3) + (0.5 / 6) U = 0.25 + 0.40 + 0.0833 **U = 0.7333 (or 73.33%)**
- **Urm(3) Value:** The theoretical Liu & Layland utilization bound for 3 tasks is calculated using the formula n(2^(1/n) - 1). Urm(3) = 3 * (2^(1/3) - 1) **Urm(3) = 0.7797 (or 77.97%)**
- **Conclusion:** Since U (0.7333) <= Urm(3) (0.7797), the task set is mathematically guaranteed to be schedulable under the Rate Monotonic scheduler without any deadline misses.

**2. What is the minimum/maximum/average response time of all tasks?** Based on the RM scheduling simulation over the hyperperiod (H = 6):

- **Task 1 (Highest Priority):** Executes at t=0, t=2, t=4. It is never preempted. Min: 0.5 | Max: 0.5 | Average: 0.5
- **Task 2 (Medium Priority):** Executes at t=0 and t=3. Its first job finishes at 1.7. Its second job is preempted by T1 at t=4 but finishes at 4.7 (response time 4.7 - 3 = 1.7). Min: 1.7 | Max: 1.7 | Average: 1.7

- **Task 3 (Lowest Priority):** Executes at t=0. It gets preempted by both T1 and T2, finishing its execution exactly at t=2.7. Min: 2.7 | Max: 2.7 | Average: 2.7

**3. Is any task missing the deadline? Which task? Where? No task is missing its deadline.** As predicted by the utilization bound check (U <= Urm(n)), all tasks successfully complete their execution well before their respective deadlines.

**4. If a deadline is missed, could it be avoided by changing the scheduler?** Since no deadlines are missed in the current RM schedule, a change of scheduler is not strictly necessary for this specific task set. However, in theory, if a task set exceeded the RM bound but stayed under U = 1.0 (100%), changing to a dynamic scheduler like **EDF (Earliest Deadline First)** could avoid deadline misses, as EDF can optimally schedule any task set up to 100% utilization.


Figure 1. (T1,T2,T3)

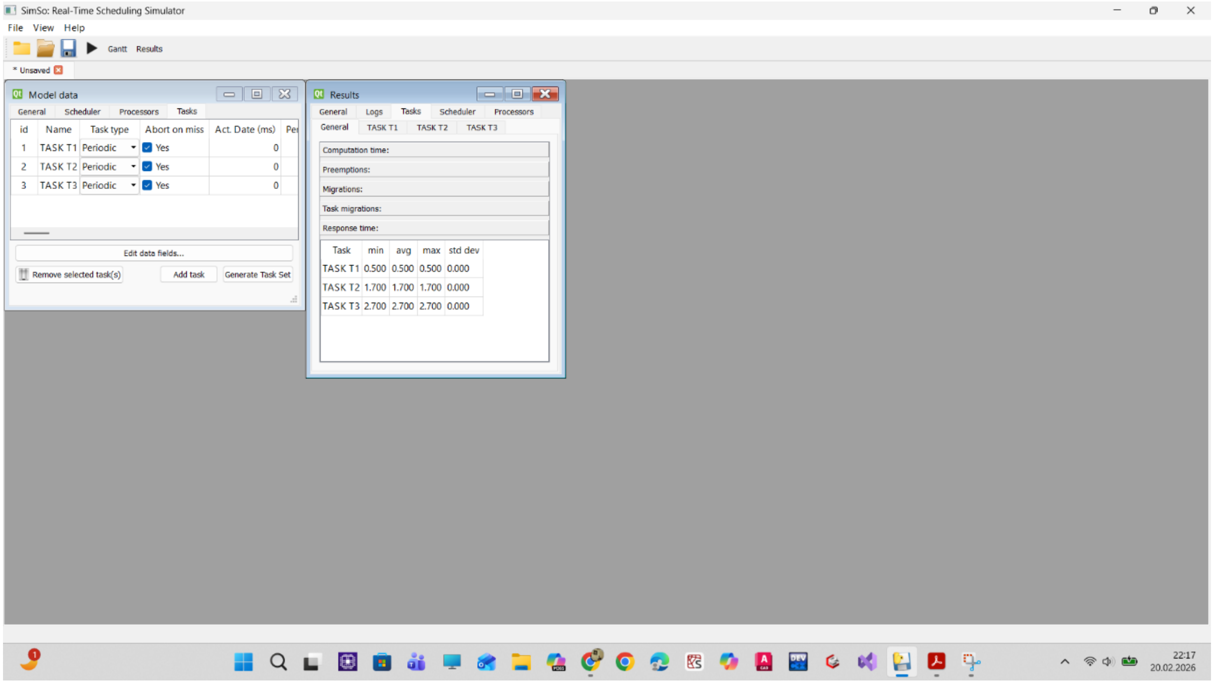

Figure 2. (Min,avg,max response times)

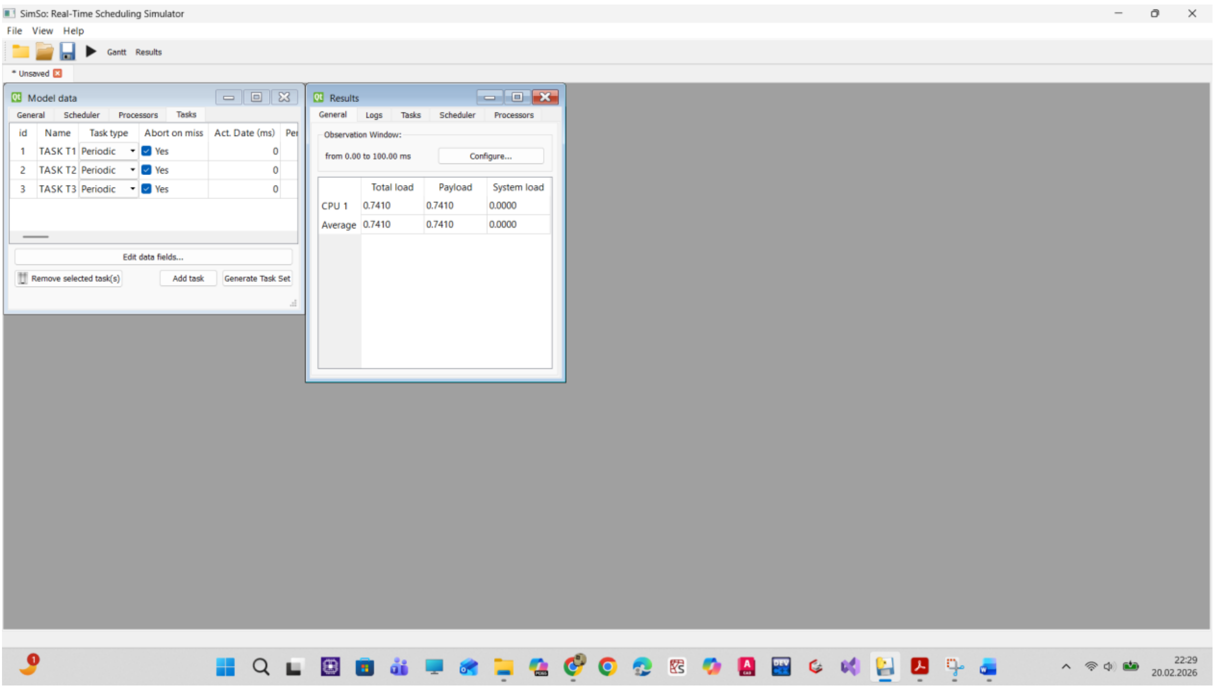

Figure 3. (Urm)

## SimSo Simulation Report: Earliest Deadline First (EDF) Scheduling

### 1. What is the utilization factor of the system and what is the value for Urm(4)?

- **System Utilization (U):** U = (0.5 / 2) + (2 / 5) + (0.1 / 1) + (5 / 10) U = 0.25 + 0.40 + 0.10 + 0.50 **U = 1.25 (125%)** *(Note: SimSo shows CPU Load as 1.00 because the physical processor limit is reached).*
- **Urm(4) Value:** Urm(4) = 4 * (2^(1/4) - 1) **Urm(4) = 0.7568 (75.68%)**

### 2. What is the minimum/maximum/average response time of all tasks?

- **System State:** Overload condition (U = 1.25 > 1.00).
- **T1 & T3 (Short deadlines):** Minimum, maximum, and average response times remain low and stable.
- **T2 & T4 (Long deadlines):** Response times are not stable. Maximum and average values continuously increase toward infinity as the simulation runs.

### 3. Is any task missing the deadline? Which task? Where?

- **Yes.** * **Which tasks:** Primarily **Task 4 (T4)**, followed by **Task 2 (T2)**.
- **Where/Why:** T4 misses deadlines starting from its early periods. Higher-priority tasks (T1, T2, T3) consume 75% of the CPU, leaving only 25% for T4, which requires 50% to finish on time.

### 4. If a deadline is missed, could it be avoided by changing the scheduler?

- **No.** * Since the total utilization demand is 125% (U = 1.25 > 1.0), the system is mathematically unschedulable on a single core. No scheduler (RM, EDF, etc.) can avoid deadline misses in an overloaded system.


Figure 4. Edf (T1,T2,T3,T4)

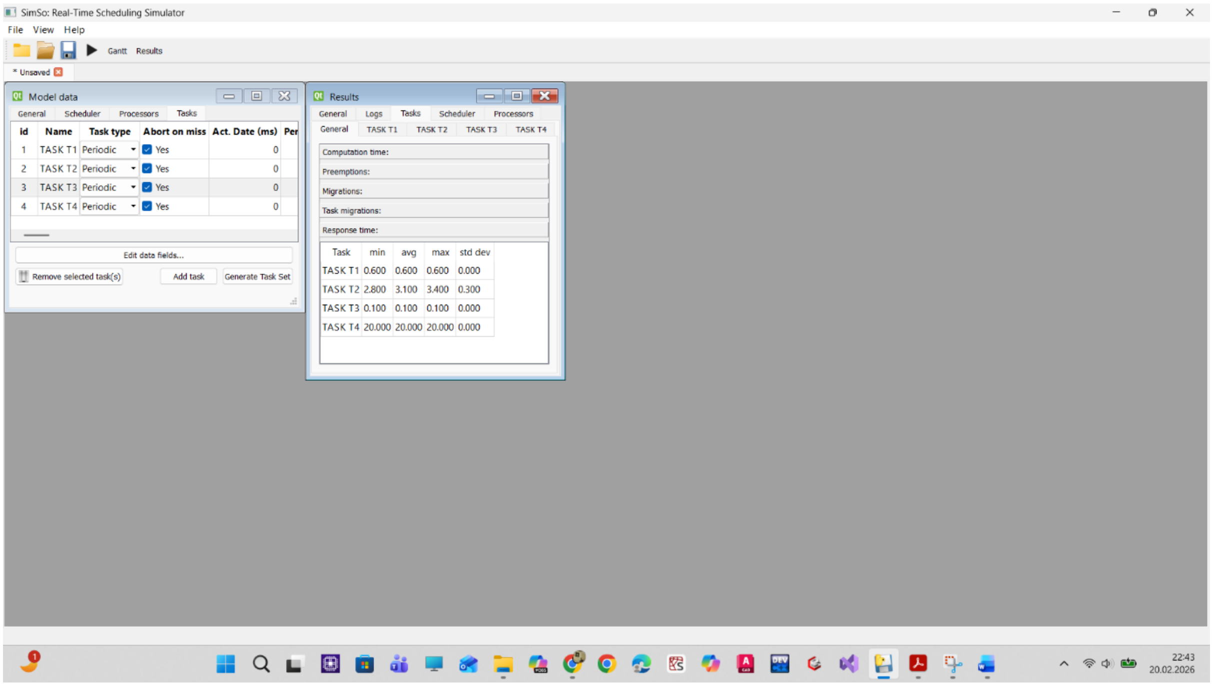


Figure 5.Edf (min,avg,max response times)

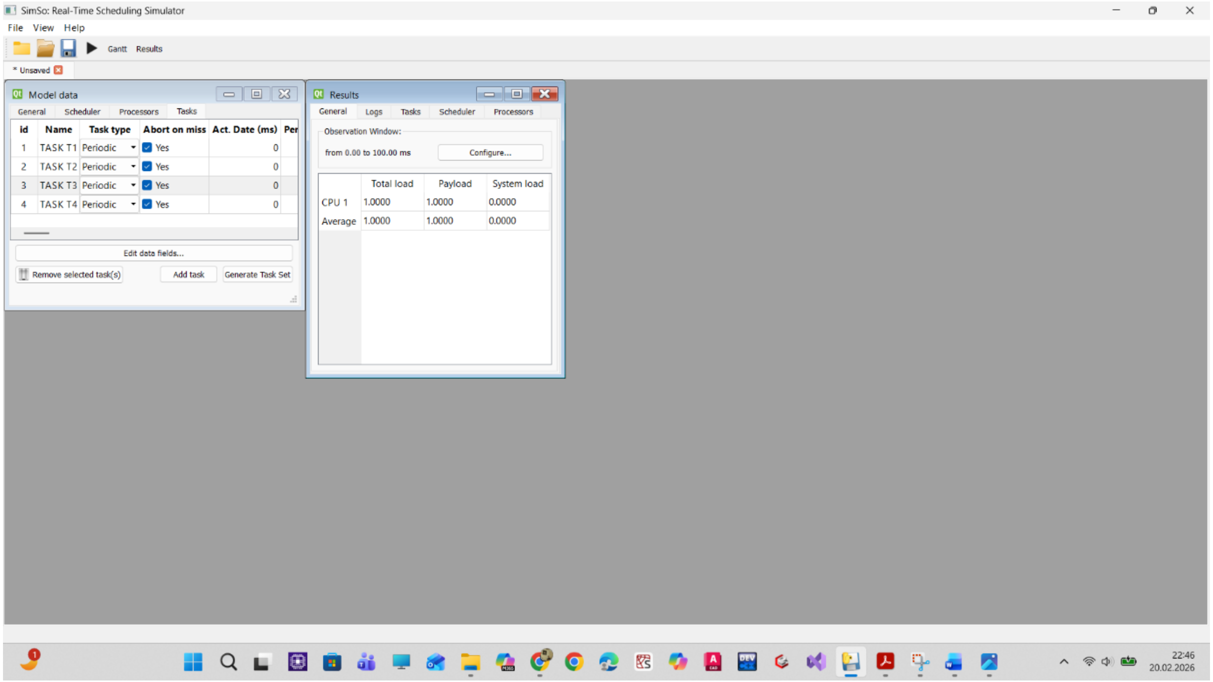


Figure 6. EDF (Urm 4)